symmetry

MDPI

Review

A Brief Survey on the Riemann Hypothesis and Some Attempts to Prove It

Renato Spigler

Department of Mathematics and Physics, Roma Tre University, 1, Largo S. Leonardo Murialdo, 00146 Rome, Italy; renato.spigler@uniroma3.it

Abstract: This paper presents a brief survey on the Riemann Hypothesis, a central conjecture in number theory with profound implications, and describes various recent attempts aimed at proving it.

Keywords: zeros of the Riemann zeta function; the Riemann Hypothesis; sum over all inverse zeta's nontrivial zeros

MSC: 11M06; 11M26

1. Introduction

The Riemann zeta function is usually defined as the Dirichlet series

$$\zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad s = \sigma + it, \tag{1}$$

[1] (Ch. 25), [2] (Ch. I, sec. 1.1, (1.1.1), p. 1), [3] (Ch.1, sec. 1.2, (1), p. 6), which converges for all $s \in \mathbf{C}$ such that $\sigma = \mathrm{Re}\{s\} > 1$, and as its meromorphic continuation in the complex plane, where it has only a simple pole at $s = 1$. It can also be defined by the Euler product formula,

$$\zeta(s) = \prod_p \left(1 - \frac{1}{p^s}\right)^{-1}, \quad \mathrm{Res} > 1,$$

where p runs through all primes, $p = 2, 3, 5, 7, \ldots$, [2] (Ch. I, sec. 1.1, (1.1.2), p. 1), [3] (sec. 2.1, (1), p. 6).

It is known that $\zeta(s)$ vanishes at the infinitely many negative even integers $s_n := -2n$, $n \in \mathbf{N}$, called the *trivial* zeros, as well as at infinitely (countably) many zeros, called *nontrivial* zeros, located on the "critical line", $\mathrm{Re}\{s\} = 1/2$, a result due to Hardy [3] (Ch. 11, sec. 11.1, p. 226)) [4] . It is not known to date whether other nontrivial zeros exist in $\mathbf{C}$, which should necessarily belong to the "critical strip", $\Sigma := \{(\sigma, t) \in \mathbf{C} : 0 < \sigma < 1, -\infty < t < +\infty\}$. That no such zeros exist is what is called the Riemann Hypothesis (RH, for short).

The importance of the RH rests on its relation with the distribution of primes [3] (Sec. 1.11, p. 22), [5], but also on many equivalent formulations, some quite technical. For some extensive lists, see [6–8], for example. There is, in addition, a large number of results whose validity would immediately follow from the validity of the RH.

check for updates

Academic Editor: Sergei Odintsov

Received: 24 December 2024
Revised: 26 January 2025
Accepted: 27 January 2025
Published: 4 February 2025

Citation: Spigler, R. A Brief Survey on the Riemann Hypothesis and Some Attempts to Prove It. *Symmetry* **2025**, *17*, 225. https://doi.org/10.3390/sym17020225

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

Indeed, the RH implies, for instance, some strong bound on the growth of the primes counting function, namely sharp estimates on the remainder term in the formulation of the prime number theorem, for which, in 1899, de la Vallée Poussin [9] proved the form

$$\pi(x) = \mathrm{Li}(x) + \mathcal{O}\left(x\, e^{-a\sqrt{\log x}}\right), \quad \text{as } x \to +\infty, \tag{2}$$

for some positive constant a, where $\mathrm{Li}(x) := \mathrm{li}(x) - \mathrm{li}(2) = \int_2^x \frac{dt}{\log t}$ is the logarithmic integral. The error term has been ameliorated in the following years. However, it is interesting that some refinements hold assuming the validity of the RH: in 1901, it was shown by Helge von Koch [10] that the RH is *equivalent* to the more precise estimate

$$\pi(x) = \mathrm{Li}(x) + \mathcal{O}\left(\sqrt{x}\,\log x\right). \tag{3}$$

A bound for the constant implied by the $\mathcal{O}$ symbol was obtained much later by Schoenfeld [11], but again under the validity of the RH.

Many results have been established over the time, the most pertaining to pure mathematics, under the condition that the RH is true. As usual, in coping with difficult unsolved mathematical problems, several collateral results are obtained. Establishing a number of new results may be even more important than achieving the final goal of proving the RH.

To study the zeros of the zeta function, the so-called (Riemann's) xi function,

$$\xi(s) := \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{4}$$

[2] (Ch. II, (2.1.12), p. 16) has also been widely used. For instance, Hardy proved that $\xi(s)$ has infinitely (countably) many zeros on the critical line $\mathrm{Re}\{s\} = 1/2$, [3] (Ch. 11, sec. 11.1, p. 226)) [4].

Numerous properties of general results about the zeta function can be found in a number of books; see [2,3,12], e.g., Several surveys, recording advances and possible paths to prove the RH have appeared regularly in the literature over the time; see, e.g., [12–16]. In particular, it should be emphasized that Titchmarsh's work, dating back to 1951 [2], is mostly useful for researchers who wish to conduct research on the RH. N. Kurokawa [17], in 2000, provided a significant body of knowledge on the RH for allied zeta-functions, but this work remains relatively unknown, partly due to its being written in Japanese. This list is not intended to be exhaustive.

The literature concerning the RH is vast and constantly growing, with several significant contributions appearing in qualified periodicals in the first half of 2024 alone. Fortunately, unlike in the case of Fermat's Last Theorem, where any amateur, regardless of his or her incompetence or lack of mathematical background, could naively attempt to tackle the problem, addressing the RH requires a solid understanding of basic mathematics, specifically, knowledge that goes beyond that of a typical sophomore.

It is claimed that some implications of the RH exist also for other fields, such as cryptography and physics. The difficult task of factoring large composite numbers, crucial for some encryption systems, could potentially benefit from a deeper description of prime number distribution. Yet, it is believed that the truth or falsity of the RH has no relevance for the development of factorization methods.

The zeta function and the distribution of primes play also a role in various models met in different branches of physics, such as classical mechanics, statistical physics, and quantum field theory (in particular, in string theory). These relations might even suggest a path towards proving the RH. Mathematics is traditionally seen as supplying the theoretical tools with which physical theories are analyzed. Here this relationship might be reversed, with quantum physics potentially offering new insights into number theory. One proposed

strategy for proving the RH is to exploit the Hilbert–Pólya conjecture (see Section 2.2). This requires determining a self-adjoint operator on a Hilbert space, whose eigenvalues would be the ordinates of the zeros of the zeta function. Since this operator is self-adjoint, its eigenvalues must be real. In quantum mechanics, systems are governed by self-adjoint operators, that is, their Hamiltonians. This analogy suggests that perhaps a quantum mechanical system could be found, whose energy levels correspond to the zeros of the zeta function.

The following set of attempts to prove the RH is far from being exhaustive. Typically, they are (recent or less recent) failures.

2. About Some Attempts to Prove the RH

Many attempts aiming either to prove or to disprove the correctness of the RH have been made since the 1859 work of Bernhard Riemann [18], resorting to a variety of methods, some of which somewhat sophisticated.

2.1. Zero-Free Regions

The interesting idea of identifying "zero-free" regions inside the critical strip should be mentioned. Over the years, there has been some interest in determining regions of the shape $\sigma \geq 1 - 1/f(|t|)$, for some function $f(t)$ tending to infinity as $t \to \pm\infty$, since this behavior has some relation to the distribution of the prime numbers.

The first of such regions was constructed in 1899 by de la Vallée–Poussin [3] (sec. 5.2, p. 79) [19], but one of the best results has been considered for long time that achieved by Vinogradov and Korobov (see the review paper [19] for references and other cases). For this reason, sometimes these regions are called "Korobov–Vinogradov regions."

In 2015, Mossinghoff and Trudgian have obtained the largest known zero-free region [20], then further improved in [21]. In [20] the authors showed that no zeros exist in the domain $\sigma \geq 1 - 1/(R_0 \log t)$, where $R_0 = 5.573412$, which result improved the previously best-known value $R_0 = 5.68371$, obtained by Jang and Kwon in 2014 [22].

Very recently, in [21] Mossinghoff, Trudgian, and Yang obtained ameliorated explicit zero-free regions. For example, no zeros of the zeta function exist in the Korobov–Vinogradov region

$$\sigma \geq 1 - \frac{1}{55.241\ (\log|t|)^{2/3}(\log\log|t|)^{1/3}} \tag{5}$$

for $|t| \geq 3$, and in the classical zero-free region

$$\sigma \geq 1 - \frac{1}{5.558691\ \log|t|} \tag{6}$$

for $|t| \geq 2$. In addition, the authors also obtained the intermediate zero-free region

$$\sigma > 1 - \frac{0.05035}{h(t)} + \frac{0.0349}{h^2(t)} \tag{7}$$

for $|t| \geq e^{1000}$, where $h(t) = \frac{27}{164}\ \log|t| + 7.096$.

Earlier, in 2006, Freitas [23] established a Li-type criterion (see Section 2.3) to obtain a necessary and sufficient condition for the existence of zero-free strips, contained inside the critical strip, $0 < \mathrm{Re}\{s\} < 1$. A numerical rigorous result due to Platt and Trudgian belongs to this kind of approach, but this will be reported later, in more detail.

2.2. The Hilbert–Pólya Conjecture

Among the most intriguing ideas proposed to prove the RH, one should perhaps recall the Hilbert–Pólya conjecture, according to which the RH is true because non-trivial zeros

of the zeta function correspond (in a certain canonical way) to the eigenvalues of some positive operator. This conjecture has been often regarded as a most promising approach. However, very little is known about its origins; some people attributed its formulation to Hilbert and Pólya, independently, some time in the 1910s.

Andrew Odlyzko had a correspondence with George Pólya between 1981–1982 [24,25], where he asked for a reference, time, and reasoning about this issue. George Pólya replied that he had spent two years in Göttingen, ending around the beginning of 1914, when he tried to learn analytic number theory from Landau. The latter asked him one day: "You know some physics. Do you know a physical reason that the Riemann hypothesis should be true". This would be the case, he answered, if the nontrivial zeros of the Xi-function were so connected with the physical problem that the Riemann hypothesis would be equivalent to the fact that all the eigenvalues of the physical problem are real. Pólya added that he never published this remark. Incidentally, the aforementioned Xi-function is defined as $\Xi(z) = \xi\left(\frac{1}{2} + zi\right)$.

That is it. It may be too little, too vague, to base some investigations on this, but several researchers, mostly physicists, took it seriously, even though still unsuccessfully.

There are also some theorems by Pólya's [26,27], [3] (Ch. 12, sec. 12.5, p. 270), that perhaps could be related to the aforementioned plan.

2.3. The Keiper–Li's Criterion

The Li's, or better, Keiper–Li's criterion, concerns the positivity of a certain numerical sequence [28–32]. In 1997, Xian-Jin Li established a necessary and sufficient condition for the validity of the RH, since then known as the "Li's criterion" [32]; see also [33]. This has the form of a set of inequalities, namely,

$$\lambda_n := \sum_{\rho}\left[1 - \left(1 - \frac{1}{\rho}\right)^n\right] \geq 0 \quad \text{for } n = 1, 2, \ldots, \tag{8}$$

the sum being extended to *all* nontrivial zeros, ρ, of the Riemann zeta function, and understood as in (13) below (Section 3). In [28], the conditions in (8) are formulated with the λ_n's strictly positive.

In some sense, there are two ways to evaluate the λ_n's. One is to look at their definition,

$$\lambda_n := \frac{1}{(n-1)!}\frac{d^n}{ds^n}\left[s^{n-1}\ln\xi(s)\right]_{s=1}, \tag{9}$$

which comes from the representation

$$\ln\xi\left(\frac{1}{1-x}\right) = \ln\xi\left(\frac{x}{x-1}\right) = -\ln 2 + \sum_{n=1}^{\infty}\lambda_n x^n$$

of the xi function (4) [33] (sec. 4, eq. (20), p. 767) [34] (sec. 4.2) . Here, the symmetry relation $\xi(s) = \xi(1-s)$ has been used. Other equivalent formulations for the λ_n's exist; see, e.g., [35], and also [30,36,37]. In several papers [30,35,36], M. Coffey derived an alternative representation of Li's λ_n coefficients and then used it to tabulate the first 25 values of λ_n (for $n = 0, \ldots, 25$) in [35], and 101 values (for $n = 0, \ldots, 100$) in [36]. These alternative coefficients involve other well-known constants, such as the Stieltjes constants and Bernoulli numbers.

In Figure 1, the behavior of λ_n is shown as a function of n, for n up to about 50. Up to $n \approx 10$, these values are very well fit by a parabola with $\lambda_n \sim 0.023 \times n^2$, then a departure from this behavior is clear [38].

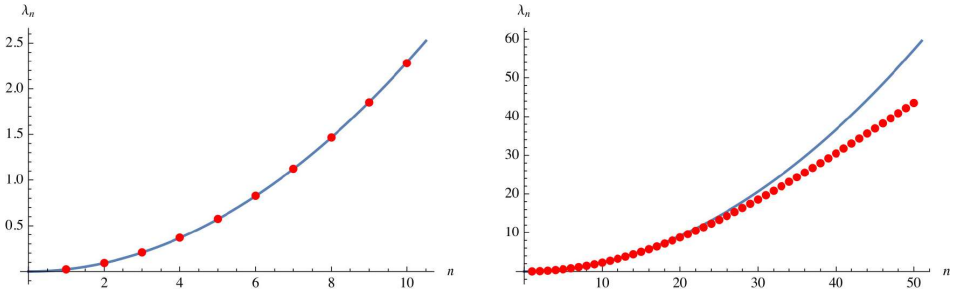


Figure 1. The coefficients λ_n vs. n.

The first 3300 Keiper–Li coefficients were evaluated by K. Maślanka in [39]. A computation for up to $n = 100{,}000$, made in [34], shows an agreement on the first two decimals with a certain "Keiper's conjecture" (hence, with the RH), providing

$$\lambda_{100,000} = 4.625807824069022314094160 38...$$

Keiper's asymptotic approximation yields in fact $\lambda_{100,000} \approx 4.626132$ [40] (pp. 63–65).

Another way to evaluate the λ_n's is using the successively obtained Formula (8); see [32]. Using the latter, however, the validity or not of the RH *cannot* be established since it involves all nontrivial zeros, without knowing whether any of them exist or not off the critical line. In [41], Voros established an alternative asymptotic behavior, basing on (8).

In order to prove (or disprove) the RH via Li's criterion, one should instead use (9), e.g., which does not involve explicitly the nontrivial zeros themselves. Trying to make explicit the Li's representation of the parameters λ_n, one could use the integral representation of $\xi^{(j)}(1)$, given in [35] (eq. (9), p. 527), and consider resorting to the celebrated Faa' di Bruno formula, which yields the n-th derivative of a given composite function. However, this formula uses sums over set partitions and hence is extremely involved, to the point that to follow this approach seems to be a hopeless task.

2.4. Horizontal Monotonicity

In [42], the so-called "horizontal behavior" of $|\zeta(s)|$, mentioned by Saidak and Zvengrowski [43], and earlier by Spira [44] was emphasized. It was observed that, at the beginning of the article on the Riemann zeta function, in the Wolfram MathWorld [45], a plot shows horizontal "ridges" of $|\zeta(\sigma + it)|$, for $0 < \sigma < 1$ and $1 < t < 100$. If indeed such ridges would decrease strictly monotonically for $0 < \sigma < 1/2$, then the RH would be proved to be true (cf. [43,46]). This behavior is easy to prove *outside* the critical strip, but nobody has been able to prove it, so far, *inside* the critical strip.

2.5. Hyperbolicity of Jensen Polynomials

In 1927, G. Pólya proved that the RH is equivalent to the so-called hyperbolicity of Jensen polynomials for the Riemann zeta function at its point of symmetry. This is called the Pólya–Jensen criterion. Advances in this direction are based in the newly discovered fact that these polynomials can be well approximated in terms of Hermite polynomials [47]. This result was initially praised by E. Bombieri [14] as a breakthrough, but this approach was soon considered actually not so useful, hence a not too promising route, by D.W. Farmer [48].

2.6. Basing on New Bounds for Large Values of Dirichlet Polynomials

Very recently, in [49], L. Guth and J. Maynard have posted a significant new result on the arXiv. Largely using Fourier analysis, the authors' advance towards the Riemann hypothesis rests on improving a result deemed insurmountable for more than 50 years, that

is, making the first substantial improvement to a classical 1940 bound of Ingham regarding the zeros of the Riemann zeta function. Field medalist Terence Tao considered this work a remarkable breakthrough. Yet, he still judged it very far from fully proving the RH.

3. Numerical Approach

A continuous progress has been made over the years in computing numerically more and more zeros *on* the critical line, and exploring the possible existence of zeros *off* it; see, e.g., [50,51]. Often, computations are based on the Riemann–Siegel formula [52]. Of course, no numerical approach can provide acceptable proofs of the RH, but the numerical path may be useful to formulate conjectures and develop insight. For instance, simulations and visualizations suggest the possible "horizontal monotonicity" of the modulus of $\zeta(s)$, even though, unfortunately, nobody has been able to prove that this occurs inside the critical strip [42].

Relevant computations of the sum of the series

$$\sum_k \frac{1}{t_k^2} \text{ and } \sum_k \frac{1}{1/4 + t_k^2},$$

$1/2 + it_k$ being the nontrivial zeros on the critical line, and estimates (with bounds) for some more general sums were carried out in [53,54]. Some attention has also been devoted to computing the sums $\sigma_k := \sum_\rho \rho^{-k}$, extended over all nontrivial zeros, for fixed k's, $k \in \mathbf{N}$. The σ_k's are also related to the Li's coefficients, λ_n, defined below; see [33] (eq. (27), p. 768), e.g.

Here are a few examples. In 1982, R.P. Brent et al. have tested the RH computationally, and confirmed its validity for the first 200,000,001 zeros [55]. Such zeros are all located in the critical strip (and lie on the critical line), up to the ordinate t = 81,702,130.19. In 2002, S. Wedeniwski has shown that the first 10^{12} nontrivial zeros lie on the critical line [56]. In [57], it is recalled that in 2004 X. Gourdon [58] used a fast method, developed by Odlyzko and Schoenhage, to verify that the first 10^{13} nontrivial zeros of the zeta function lie on the critical line, which fact validates the RH up to about the ordinate 2.4×10^{12}, in the critical strip; see [59]. Actually, A.M. Odlyzko in [51,60] showed the heights of the zeros of the Riemann zeta function numbered $10^{22} + 1$ up to $10^{22} + 10^4$.

More precisely, in 1988, Odlyzko and Schoenhage developed a fast algorithm for evaluating the Riemann zeta function at many points, using the Fast Fourier Transform [61]. This algorithm reduces the number of operations required to evaluate a finite Dirichlet series of length N at $\mathcal{O}(N)$ equally spaced values from $\mathcal{O}(N^2)$ to $\mathcal{O}(N^{1+\varepsilon})$. Such a method would enable testing the RH for the first n zeros in $\mathcal{O}(n^{1+\varepsilon})$ operations, as opposed to $\mathcal{O}(n^{3/2})$ operations required by earlier methods, provided that no multiple zeros or closely spaced zeros of the zeta function occur.

Later, in 2001 Odlyzko computed the first 10 billion zeros of the Riemann zeta function near height 10^{22}, thus verifying the RH for these zeros and providing further evidence for other conjectures related to the distribution of zeros of the Riemann zeta function [60].

In his 2004 paper, Gourdon presented an optimization of the Odlyzko–Schoenhage algorithm [58]. This efficiently computes the Riemann zeta function at large heights on the critical line and facilitates the computation of zeros of the zeta function through its implementation. He was able to compute two billion zeros from the 1024-th zero of the Riemann Zeta function.

More recently, in 2021, Platt and Trudgian established a remarkable result, verifying that the RH is true up to the height 3×10^{12}, in a rigorous way, using *interval arithmetic* [62]. In this way, numerical results can provide proofs.

As it is well known, in order to prove the RH, it suffices to show that no zero of the zeta function exists in the *semi-infinite open critical* upper *half-strip*, $H := \{(\sigma, t) \in \mathbf{C} : 1/2 <$

$\sigma < 1, t > 0\}$. In fact, if a nontrivial zero, say ρ, exists in Σ, then also its conjugate, $\overline{\rho}$, would be a zero (since $\zeta(s)$ is meromorphic and real on the real line), as well as its symmetric one with respect to the critical line $\mathrm{Re}\{s\} = 1/2$, that is $1 - \rho$, and thus $1 - \overline{\rho}$ as well. This follows from the functional equation obeyed by the zeta function,

$$\zeta(s) = 2^s \pi^{s-1} \sin\Big(\frac{\pi}{2} s\Big) \Gamma(1-s)\, \zeta(1-s) \tag{10}$$

[2] (Ch. II, (2.1.1), p. 13), [3] (eq. (4), p. 13), also written, changing s into $1 - s$, $\zeta(1-s) = 2^{1-s}\pi^{-s}\cos\big(\frac{\pi}{2}s\big)\Gamma(s)\zeta(s)$ [2] (Ch. II, (2.1.8), p. 16).

It follows that *nontrivial zeros would come in groups of four* if they exist in the critical strip, off the critical line, and in pairs if they are on the critical line. Therefore, if $s_0 := \sigma_0 + it_0$ with $1/2 < \sigma_0 < 1$ and $t_0 > 0$ is a zero, the Weierstrass factorization of the entire function ξ would contain terms like

$$\{(s - s_0)(s - \bar{s_0})[s - (1 - s_0)][s - (1 - \bar{s_0})]\}^{m_0}, \tag{11}$$

where $m_0 \in \mathbf{N}$ would be the multiplicity of s_0.

On the other hand, it is common knowledge too that no zeros exist on the intersection of the critical strip and the real line, as well as on the lines $\mathrm{Re}\{s\} = 0$, $\mathrm{Re}\{s\} = 1$. One can thus confine the discussion to the set H, but this region can be restricted further. In fact, it is also known that no nontrivial zero exists in H with an ordinate less than some value. For instance, in [63] (Ex. 10.2.1, (c), p. 353) it is recorded that, if $\rho_0 = \sigma_0 + it_0$ is a nontrivial zero with $\sigma_0 \neq 1/2$, then $|t_0| > 9.2518^+$.

However, as mentioned above, in a recent paper, Platt and Trudgian have shown numerically, yet *rigorously*, using interval arithmetic, that the RH is true up to about the height 3×10^{12} (cf. [58,59]), that is, all zeros existing in the critical strip, with imaginary part $0 < t \leq 3 \times 10^{12}$, have real part $\mathrm{Re}\{s\} = 1/2$ (and are all simple), [62]. Consequently, any investigation aimed at ruling out the existence of nontrivial zeros can be confined just to the set $H_{t^*} := H \cap \{t > t^*\}$, for some t^*, that can be taken equal to 3×10^{12}. Considering the shape of the known zero-free regions, we can state that any hypothetical nontrivial zero should be located in such set H_{t^*} and very close to and at the right of the vertical line $\mathrm{Re}s = 1/2$, or very close to and at the left of the vertical line $\mathrm{Re}s = 1$ (or symmetrically, of course, both with respect the line $\mathrm{Re}s = 1/2$ and to the real line).

That no nontrivial zeros exist off the critical line, below some value of t can also be shown aby pplying the well-known equation for the sum of the inverse nontrivial zeta's zeros,

$$\sum_{\rho} \frac{1}{\rho} = 1 + \frac{\gamma}{2} - \frac{1}{2}\ln(4\pi) = 0.0230957^+, \tag{12}$$

where the sum is extended over *all* nontrivial zeros of the zeta function, and $\gamma = 0.5772156^+$ is the Euler–Mascheroni constant. This equation was already known to Riemann [3] (Ch. 3, sec. 3.8, p. 67, eq. (4)), and appears also in Li's criterion [28,31,32]. The sum on the left-hand side of (12) does not converge absolutely, but each term $1/\rho$ is intended to be grouped with its conjugate [64] (Ch. 12, p. 81), [65] (p. 214), and the sum is understood as

$$\sum_{\rho} := \lim_{T \to +\infty} \sum_{|\mathrm{Im}\{\rho\}| \leq T} \tag{13}$$

termed "$*$-convergence" by Bombieri and Lagarias [who also used the unusual notation x^+ to denote chopping off further decimals of x] [28] (p. 275). As already recalled, each nontrivial zero is necessarily accompanied by its complex conjugate as well as by its symmetric one with respect to the critical line (if it lies off this line), and thus by the

conjugate of the latter too. Therefore, as we stated above, if a nontrivial zero, say $\rho_0 = \sigma_0 + it_0$, exists in the critical strip, off the critical line, with $1/2 < \sigma_0 < 1$, $t_0 > 0$, then also $\overline{\rho}_0$, $1 - \rho_0$, and $1 - \overline{\rho}_0$ will contribute to the sum on the left-hand side of (12) with the amount

$$\frac{1}{\rho_0} + \frac{1}{\overline{\rho}_0} + \frac{1}{1-\rho_0} + \frac{1}{1-\overline{\rho}_0}$$

$$= \frac{2\mathrm{Re}\{\rho_0\}}{|\rho_0|^2} + \frac{2(1-\mathrm{Re}\{\rho_0\})}{|1-\rho_0|^2} = 2\left(\frac{\sigma_0}{\sigma_0^2+t_0^2} + \frac{1-\sigma_0}{(1-\sigma_0)^2+t_0^2}\right) \geq \frac{2}{1+t_0^2}. \tag{14}$$

For the last inequality, the result of [63] (Ex. 10.2, 8. (a), p. 355) was used. This estimate is useful to rule out that some ρ_0 might be a nontrivial zero.

In fact, if we define the following sums of the inverses of zeros:

- S_{all}: when extended to *all* nontrivial zeros (including both those on the critical line and those off the critical line, if any exist), see Equation (12);
- S_{on}: when extended only to all nontrivial zeros located on the critical line;
- S_{off}: when extended only to all nontrivial zeros off the critical line;
- $S_{\mathrm{on,200}}$: when extended only to the first 200 nontrivial zeros located on the critical line (note that each zero with $t > 0$ is included along with its conjugate),

then we have

$$S_{\mathrm{all}} = S_{\mathrm{on}} + S_{\mathrm{off}} \geq S_{\mathrm{on,200}} + \frac{2}{1+t_0^2} \tag{15}$$

since $S_{\mathrm{on,200}}$ does not include all nontrivial zeros lying on the critical line, and $\frac{2}{1+t_0^2}$ represents the contribution of a single nontrivial zero located off the critical line (along with its complex conjugate, its symmetrical counterpart, and the conjugate of the symmetrical counterpart). Since

$$S_{\mathrm{all}} = 1 + \frac{\gamma}{2} - \frac{1}{2}\ln(4\pi) \approx 0.0230957 \tag{16}$$

by (12), and

$$S_{\mathrm{on,200}} \approx 0.021, \tag{17}$$

as reported in [66] (p. 249), it follows that

$$\frac{2}{1+t_0^2} \leq S_{\mathrm{all}} - S_{\mathrm{on,200}} \approx 0.0230957 - 0.021 = 0.0020957, \tag{18}$$

which implies the approximate bound

$$t_0 \geq 30.918. \tag{19}$$

That is, if a nontrivial zero like $\sigma_0 + it_0$ exists off the critical line, then, necessarily, t_0 must be larger than roughly 30.918. No such zero can exist below this ordinate.

Of course, significantly larger lower bounds are known to exist: recall for instance, the various results on zero-free regions or the Platt and Trudgian bound of 3×10^{12}.

4. Summary

It has long been known that *all* nontrivial zeros of the Riemann zeta function must lie within the open critical strip $\Sigma := \{(\sigma, t) \in \mathbf{C} : 0 < \sigma < 1, -\infty < t < +\infty\}$. Although the existence of infinitely (countably) many zeros on the critical line, $\mathrm{Re}\{s\} = 1/2$, has been known since 1914, no proof of the existence or nonexistence of other nontrivial zeros within Σ has been found to date. The Riemann Hypothesis (RH) states that no such other

nontrivial zeros exist. This short survey describes a few more or less recent attempts to prove the RH.

Funding: This research received no external funding.

Acknowledgments: The author expresses gratitude to Richard Brent, David Platt, Timothy Trudgian, Purusottum Rath, and Sanoli Gun for valuable private communications. This work was conducted within the framework of the Italian GNFM-INdAM.

Conflicts of Interest: The author declares no conflicts of interest.

References

1. NIST Digital Library of Mathematical Functions. Available online: https://dlmf.nist.gov/ (accessed on 23 December 2024).
2. Titchmarsh, E.C. *The Theory of the Riemann Zeta-Function*, 2nd ed.; Heath-Brown, D.R., Ed.; Clarendon Press: Oxford, UK, 1986.
3. Edwards, H.M. *Riemann's Zeta Function*; Academic Press: New York, NY, USA,1974.
4. Hardy, G.H. Sur les zéros de la fonction $\xi(s)$ de Riemann. *C. R. Acad. Sci. Paris* **1914**, *158*, 1012–1014.
5. Jameson, G.J.O. *The Prime Number Theorem*; First Published: 2003; Cambridge University Press: Cambridge, UK, 2004;
6. The Riemann Hypothesis. Available online: https://www.aimath.org/WWN/rh/rh.pdf (accessed on 23 December 2024).
7. Number theory and physics archive. Available online: https://empslocal.ex.ac.uk/people/staff/mrwatkin/zeta/physics.htm (accessed on 23 December 2024).
8. Riemann hypothesis. Available online: https://en.wikipedia.org/wiki/Riemann_hypothesis (accessed on 23 December 2024).
9. de la Vallée Poussin, C.-J. Sur la fonction $\zeta(s)$ de Riemann et le nombre des nombres premiers inférieurs a une limite donnée. *Mémoires Couronnés l'Académie Belg.* **1899**, *59*, 1–74.
10. von Koch, H. Sur la distribution des nombres premiers [On the distribution of prime numbers]. *Acta Math.* **1901**, *24*, 159–182. (In French) [CrossRef]
11. Schoenfeld, L. Sharper bounds for the Chebyshev functions $\theta(x)$ and $\psi(x)$, II. *Math. Comp.* **1976**, *30*, 337–360. [CrossRef]
12. Borwein, P.; Choi, S.; Rooney, B.; Weirathmueller, A. *The Riemann Hypothesis: A Resource for the Afficionado and Virtuoso Alike*; Spring, CMS Books in Mathematics; Springer: Berlin, Germany, 2007; ISBN 978-0-387-72125-5. [CrossRef]
13. Bombieri, E. The classical theory of zeta and L-functions. *Milan J. Math.* **2010**, *78*, 11–59. [CrossRef]
14. Bombieri, E. New progress on the zeta function: from old conjectures to a major breakthrough. *Proc. Natl. Acad. Sci. USA* **2019**, *116*, 11085–11086. [CrossRef]
15. Conrey, J.B. The Riemann Hypothesis. *Notices Am. Math. Soc.* **2003**, *50*, 341–353.
16. Sarnak, P. Problems of the Millennium: The Riemann Hypothesis. 2004. Available online: https://www.claymath.org/sites/default/files/sarnak_rh_0.pdf (accessed on 23 December 2024).
17. Kurokawa, N. *The 15 years of the Riemann Hypothesis*; Iwanami-shoten: Tokyo, Japan, 2000. (In Japanese)
18. Riemann, B. Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse. *Monatsberichte Berl. Akad.* **1859**, *2*, 2.
19. Ford, K. Zero-free regions for the Riemann zeta function. In *Number Theory for the Millennium, II (Urbana, IL, 2000)*; A.K. Peters: Natick, MA, USA, 2002; pp. 25–56.
20. Mossinghoff, M.J.; Trudgian, T.S. Nonnegative trigonometric polynomials and a zero-free region for the Riemann zeta-function. *J. Number Theory* **2015**, *157*, 329–349. [CrossRef]
21. Mossinghoff, M.J.; Trudgian, T.S.; Andrew Yang, A. Explicit zero-free regions for the Riemann zeta-function. *Res. Number Theory* **2024**, *10*, 11. [CrossRef]
22. Jang, W.J.; Kwon, S.H. A note on Kadiri's explicit zero free region for Riemann zeta function. *J. Korean Math. Soc.* **2014**, *51*, 1291–1304. [CrossRef]
23. Freitas, P. A Li-type criterion for zero-free half-planes of Riemann's zeta function. *J. Lond. Math. Soc.* **2006**, *73*, 399–414. [CrossRef]
24. Andrew Odlyzko: Tables of zeros of the Riemann zeta function. Available online: https://www-users.cse.umn.edu/~odlyzko/zeta_tables/index.html (accessed on 23 December 2024).
25. Hilbert–Pólya conjecture. Available online: https://en.wikipedia.org/wiki/Hilbert%E2%80%93P%C3%B3lya_conjecture (accessed on 23 December 2024).
26. Pólya, G. Über die Nullstellen gewisser ganzer Funktionen. *Math. Z.* **1918**, *2*, 352–383. [CrossRef]
27. Pólya, G. Über trigonometrische Integrale mit nur reellen Nullstellen. *J. Reine Angew. Math.* **1927**, *158*, 6–18. [CrossRef]
28. Bombieri, E.; Lagarias, J. Complements to Li's criterion for the Riemann hypothesis. *J. Number Theory* **1999**, *77*, 274–287. [CrossRef]
29. Brown, F. Li's criterion and zero-free regions of L-functions. *J. Number Theory* **2005**, *111*, 1–32. [CrossRef]
30. Coffey, M.W. Toward verification of the Riemann hypothesis: Application of the Li criterion. *Math. Phys. Anal. Geom.* **2005**, *8*, 211–255. [CrossRef]
31. Gun, S.; Murty, M.R.; Rath, P. Transcendental sums related to the zeros of zeta functions. *Mathematika* **2018**, *64*, 875–897. [CrossRef]

32. Li, X.-J. The positivity of a sequence of numbers and the Riemann hypothesis. *J. Number Theory* **1997**, *65*, 325–333. [CrossRef]
33. Keiper, J.B. Power series expansions of Riemann's ξ function. *Math. Comp.* **1992**, *58*, 765–773.
34. Johansson, F. Rigorous high-precision computation of the Hurwitz zeta function and its derivatives. *Numer. Algorithms* **2015**, *69*, 253–270. [CrossRef]
35. Coffey, M.W. Relations and positivity results for the derivatives of the Riemann ξ function. *J. Comput. Appl. Math.* **2004**, *166*, 525–534. [CrossRef]
36. Coffey, M.W. New results concerning power series expansions of the Riemann xi function and the Li/Keiper constants. *Proc. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci.* **2008**, *464*, 711–731. [CrossRef]
37. Matiyasevich, Y.V. A relationship between certain sums over trivial and nontrivial zeros of the Riemann zeta-function. *Math. Notes Acad. Sci. USSR* **1989**, *45*, 131–135. [CrossRef]
38. Wolfram MathWorld. Available online: https://mathworld.wolfram.com/LisCriterion.html (accessed on 23 December 2024).
39. Maślanka, K. Li's criterion for the Riemann hypothesis–numerical approach. *Opusc. Math.* **2004**, *24*, 103–114.
40. Johansson, F. Fast and Rigorous Computation of Special Functions to High Precision. Ph.D. Thesis, Johannes Kepler Universität, Linz, Austria, 2014.
41. Voros, A. Sharpening of Li's criterion for the Riemann hypothesis. *Math. Phys. Anal. Geom.* **2006**, *9*, 53–63. [CrossRef]
42. Matiyasevich, Y.; Saidak, F.; Zvengrowski, P. Horizontal monotonicity of the modulus of the zeta function, the *L*-function, and related functions. *Acta Arith.* **2014**, *166*, 189–200. [CrossRef]
43. Saidak, F.; Zvengrowski, P. On the modulus of the Riemann zeta function in the critical strip. *Math. Slovaca* **2003**, *53*, 145–172.
44. Spira, R. An inequality for the Riemann zeta-function. *Duke Math. J.* **1965**, *32*, 247–250. [CrossRef]
45. Wolfram MathWorld. Available online: http://mathworld.wolfram.com/RiemannZetaFunction.html (accessed on 23 December 2024).
46. Borwein, J.; Bailey, D. *Mathematics by Experiment–Plausible Reasoning in the Twenty-First Century*; A. K. Peters: Wellesley, MA, USA, 2003.
47. Griffin, M.; Ono, K.; Rolen, L.; Zagier, D. Jensen polynomials for the Riemann zeta function and other sequences. *Proc. Natl. Acad. Sci. USA* **2019**, 116, 11103–11110. [CrossRef]
48. Farmer, D.W. Jensen polynomials are not a plausible route to proving the Riemann Hypothesis. *Adv. Math.* **2022**, *411*, 108781. [CrossRef]
49. Guth, L.; Maynard, J. New large value estimates for Dirichlet polynomials. *arXiv* **2024**, arXiv:2405.20552v1.
50. LMFDB, Database of L-Functions, Modular Forms, and Related Objects. Available online: https://www.lmfdb.org/zeros/zeta/ (accessed on 23 December 2024).
51. Odlyzko, A. Tables of Zeros of the Riemann Zeta Function. Available online: http://www.dtc.umn.edu/~odlyzko/zeta_tables/index.html (accessed on 23 December 2024).
52. Siegel, C.L. Über Riemanns Nachlaßzur analytischen Zahlen-theorie. *Quellen Studien Geschichte Math. Astr. Phys.* **1932**, *2*, 45–80.
53. Brent, R.P.; Platt, D.J.; Trudgian, T. Accurate estimation of sums over zeros of the Riemann zeta-function. *Math. Comp.* **2021**, *90*, 2923–2935. [CrossRef]
54. Mossinghoff, M.J.; Trudgian, T.S. Oscillations in the Goldbach conjecture. *J. Théor. Nombres Bordeaux* **2022**, *34*, 295–307. [CrossRef]
55. Brent, R.P.; van de Lune, J.; te Riele, H.J.J.; Winter, D.T. On the zeros of the Riemann zeta function in the critical strip. II. *Math. Comp.* **1982**, *39*, 681–688. [CrossRef]
56. ZetaGrid Homepage. Available online: http://www.zetagrid.net/ (accessed on 23 December 2024).
57. Voros, A. Discretized Keiper/Li approach to the Riemann hypothesis. *Exp. Math.* **2020**, *29*, 452–469. [CrossRef]
58. Gourdon, X. The 10^{13} First Zeros of the Riemann Zeta Function, and Zeros Computation at Very Large Height, Preprint, 2004. Available online: http://numbers.computation.free.fr/Constants/Miscellaneous/zetazeros1e13-1e24.pdf (accessed on 23 December 2024).
59. Goodman, L.; Weisstein, E.W. "Riemann Hypothesis." From MathWorld—A Wolfram Web Resource. Available online: https://mathworld.wolfram.com/RiemannHypothesis.html (accessed on 23 December 2024).
60. Odlyzko, A.M. The 10^{22}-nd zero of the Riemann zeta function. *Dynamical, spectral, and arithmetic zeta functions (San Antonio, TX, 1999)*, pp. 139–144. *Contemp. Math.* **2001**, *290*.
61. Odlyzko, A.M.; Schönhage, A. Fast algorithms for multiple evaluations of the Riemann zeta function. *Trans. Amer. Math. Soc.* **1988**, *309*, 797–809. [CrossRef]
62. Platt, D.; Trudgian, T. The Riemann hypothesis is true up to $3 \cdot 10^{12}$. *Bull. Lond. Math. Soc.* **2021**, *53*, 792–797. [CrossRef]
63. Montgomery, H.L.; Vaughan, R.C. *Multiplicative Number Theory: I. Classical Theory*; Cambridge University Press: New York, NY, USA, 2006.
64. Davenport, H. *Multiplicative Number Theory*, 2nd ed.; first ed.: 1967; Springer: Berlin/Heidelberg, Germany, 1980.

65. Bach, E.; Shallit, J. *Algorithmic Number Theory, Vol. I: Efficient Algorithms*; The MIT Press: Cambridge, MA, USA; London, UK, 1996.
66. Borwein, J.M.; Bradley, D.M.; Crandall, R.E. Computational strategies for the Riemann zeta function. *J. Comput. Appl. Math.* **2000**, *121*, 247–296. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.